# Kaustav Purkayastha

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GitHub | LinkedIn | Portfolio

## Skills

**Languages & Data:** SQL, PL/SQL, Redshift SQL, Shell Scripting, Python, PySpark; ETL/ELT, CDC, SCD Type-1 & Type-2, Pushdown Optimization, Data Warehousing, Data Quality & Validation, Source-to-Target Reconciliation

**Cloud & Platforms:** AWS (Redshift, S3, Glue, EC2, Aurora), Databricks, Delta Lake, Snowflake, dbt, Informatica IDMC/IICS, MS SQL Server, PostgreSQL, Oracle

**Tools & DevOps:** Stonebranch, Git/GitHub, GitHub Actions, Jenkins, Liquibase, Docker, Jira (Agile/Scrum), Apache Airflow, Amazon Bedrock

## Work Experience

**Deloitte Consulting, Kolkata** Oct 2023 – Present

***Data Engineer – AI & Data Practice***

- Building **gold-layer (mart) dimensional models** on **Delta Lake** as part of a live migration from the legacy Redshift warehouse to a **Databricks medallion lakehouse**, sourcing curated data from the lake and using **dbt** ref/source dependencies to structure the transformation DAG.
- Converting legacy **IICS ETL jobs** into **dbt ELT models** against the lakehouse, using an **AI-assisted conversion**, debugging, and reconciliation framework to validate parity between legacy and re-platformed outputs.
- Architected **70+ Informatica IDMC ETL pipelines** to retire a legacy on-prem reporting warehouse for a Fortune 500 insurer, integrating enterprise sources (SQL Server, mainframe, valuation systems) and loading 35+ years of history into **60+ Amazon Redshift tables** using **CDC-based ingestion** and **SCD Type 1/Type 2** load logic.
- Enabled accurate point-in-time and historical reporting by designing **multi-grain historization logic** (yearly, monthly, and point-in-time snapshots) for the Redshift dimensional model, reverse-engineering record effective dates from an undocumented main-and-archive source while preserving natural keys and audit integrity.
- Consolidated 4+ legacy Tableau Prep workflows into scalable **Redshift ELT pipelines** validated to **~98% accuracy** against legacy business extracts, by unifying multiple RDBMS, external, and flat-file (CSV) sources and re-engineering Python annuity calculations into optimized **set-based SQL**.
- Eliminated a Dremio dependency by migrating interdependent **semantic views to Redshift materialized views**, resolving hierarchical dependencies and performing **source-to-target reconciliation** to confirm data parity.
- Built a **metadata-driven looping framework** in Informatica IDMC for sequential SCD-preserving mainframe file ingestion via **Stonebranch**, and converted compute-intensive IICS jobs into **Redshift ELT pushdown** (post-SQL in-warehouse), accelerating conversion using Claude via **Amazon Bedrock**.

## Personal Projects

**HealthPipe AI** *View Source Code* | *View Demo* Mar 2026 – May 2026

Flask workbench that onboards healthcare datasets through a **four-agent pipeline**, surfacing **quality scorecards**, AI-generated **data dictionaries**, **NL→SQL** querying, and a prebuilt **US States Health Mart**; on-device **Gemma 3 4B**, optional **Gemini Flash**. Built as part of Deloitte's Accelerate with AI — In-person Engineering Lab Session.

*Python, Flask, DuckDB, pandas, Ollama, Pytest, GitHub Actions*

**TestNucleus** *View Source Code* Jun 2025 – Aug 2025

Config-driven **data quality framework** that separates test definitions (**YAML**) from execution logic. Supports **20+ validation checks** across completeness, uniqueness, and referential integrity — built for **CI/CD and pipeline integration**.

*Python, SQLAlchemy, Pydantic, Pytest*

## Education

**Lovely Professional University** 2024 – 2026
Master of Computer Applications, Data Science & Finance *CGPA: 7.48*

**Assam University** 2020 – 2023
Bachelor of Science with Honours, Computer Science *CGPA: 8.86*

## Certifications & Recognition

- **Certifications:** Snowflake SnowPro Core (2026), Databricks Certified Data Engineer Associate (2025), Claude Certified Associate – Foundations (2026), Industry Proficiency: Financial Services (2025), IDMC Associate (2024)

- **Recognition:** Outstanding Performance Award (Feb 2026, May 2025) | Applause Award (Jul 2024) | FY24 Rising Stars (May 2024) | Institutional Merit Scholarship (Dec 2024)